Dhruv Bhardwaj

Backend Software Engineer | New Grad
TypeScript | Python | Java | PostgreSQL | Distributed Systems
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Education

University of British Columbia *May 2026*
B.Sc. in Computer Science
Certification: Deep Learning Specialization, DeepLearning.AI

Technical Skills

Languages: Java, Python, TypeScript, C#, SQL
Backend: Node.js, FastAPI, REST APIs, tRPC, Prisma, WebSockets, RBAC, API Design
Databases: PostgreSQL, Relational Modeling, Query Optimization, ETL
Systems & Cloud: Microservices, Distributed Systems, Event Streaming, Kafka, GCP Load Balancing
Tools: Docker, Git, Linux, CI/CD, Alembic, Load Testing
ML/AI: OpenAI API, LLM Routing, RAG, TensorFlow, Keras, MobileNetV2, CRNN OCR, TensorFlow Lite

Experience

Backend Engineer | *Astin Analytics* *May 2026 – Present*

- Auditing and re-architecting a legacy Python/PostgreSQL backend to identify scalability, security, and maintainability risks across core services.
- Designing a security classification layer for a legacy RAG pipeline to sanitize inputs, isolate retrieval paths, and reduce unsafe prompt-to-context behavior.
- Prototyped Kafka-based event-streaming workflows and validated GCP load-balancing behavior through load-testing experiments for availability-focused deployment planning.

Software Engineer | *Visualization & Emerging Media Studio, UBC* *Sept 2024 – Apr 2026*

- Engineered real-time distributed backend services for multi-user XR scientific visualization, supporting cross-device state synchronization across live demo environments.
- Reduced interaction latency to less than 80 ms by optimizing WebSocket data pipelines and structured input streams for low-latency XR control.
- Improved stability for live XR visualization sessions used by 150+ students, researchers, and visiting university stakeholders by resolving networking and synchronization bottlenecks.

Backend Lead | *University Exam Generation and Analytics Platform, UBC* *May 2025 – Sept 2025*

- Architected a microservice backend using Node.js, TypeScript, tRPC, Prisma, and PostgreSQL to support exam generation, OMR grading, analytics, and result review workflows.
- Automated grading and analytics for 1000+ OMR submissions per exam cycle by building CSV ingestion, answer parsing, normalized result storage, and reporting APIs.
- Designed immutable snapshot-based grading to preserve published exam content, variant answer keys, and historical student results for reproducible grade audits.
- Secured multi-tenant data with JWT-based RBAC and PostgreSQL Row-Level Security while adding cheating-detection logic for same-variant and cross-variant answer similarity.

Developer | *GitHub Analytics Research Project, UBC* *Jan 2026 – Apr 2026*

- Extended an existing Python analytics codebase to process activity across 25 GitHub repositories, including 20K+ commits, 7K+ pull requests, and 8 months of contribution data.
- Built data-cleaning and aggregation workflows to analyze commits, pull requests, reviews, contributors, and noisy activity patterns across multi-repository datasets.

Selected Projects

Reflect AI – Safety-Aware LLM Companion App
Python | FastAPI | OpenAI API | JWT | SQLite | React Native | Expo

- Built a safety-aware LLM execution pipeline with risk classification, response-mode routing, deterministic crisis bypass, and OpenAI-backed response generation.
- Developed authenticated conversation APIs with JWT auth, async database sessions, Alembic migrations, and persistent message/safety-event logging.

PillCare – Multi-Modal Pill Identification System
TensorFlow | Keras | MobileNetV2 | CRNN OCR | OpenFDA | TensorFlow Lite

- Built a multi-stage pill identification pipeline combining MobileNetV2 visual classification, CRNN-based imprint OCR, and OpenFDA metadata retrieval across 16 drug classes.
- Improved test accuracy from 80% to 84% with gated vision-OCR fusion, class-weighted training, and targeted augmentation on a 707-image pill dataset.
- Optimized the trained model for edge deployment using TensorFlow Lite to support future mobile medication-identification workflows.